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Optimization and performance analysis of UAV-based full-duplex systems with residual transceiver hardware impairments

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ABSTRACT

In this paper, we study a full-duplex (FD) unmanned aerial vehicle (UAV) communication system affected by residual transceiver hardware impairments (RTHI) and residual self-interference (RSI). We derive analytical expressions for the symbol error probability (SEP) and achievable rate (AR) under the separate and combined effects of RTHI and RSI. To mitigate the negative impact of these impairments, particularly RSI inherent in FD operation, we propose an optimal UAV transmit power scheme that minimizes the SEP or, equivalently, maximizes the AR, subject to RTHI, RSI, and a practical transmit power constraint. The derived expressions are validated through Monte-Carlo simulations. Numerical results show that both RTHI and RSI significantly affect system performance. When one impairment is small, the effect of the other becomes more noticeable. When both are strong, SEP and AR rapidly saturate. In such cases, the proposed power control strategy helps reduce SEP, increase AR, and prevent saturation at high transmit power levels. The results also show that using optimal transmit power improves the performance of FD systems compared to conventional half-duplex (HD) systems. Importantly, the optimal transmit power is much lower than the traditional power commonly used by UAVs. As a result, the full-duplex unmanned aerial vehicle (FD-UAV) system with RTHI consumes less power than the scenario without power optimization. We also analyze the impact of modulation order, bandwidth, UAV altitude, and operating frequency on SEP and AR. Based on the results, we provide useful recommendations to improve the performance of FD-UAV systems under RTHI and RSI.

1. Introduction

In recent years, unmanned aerial vehicle (UAV) communication has become a vital component of modern wireless networks due to its flexibility, mobility, and rapid deployment capabilities [1]. UAVs can function as aerial base stations, relays, or mobile users to improve coverage and capacity in areas with limited infrastructure, such as disaster zones, rural regions, or large-scale public events [1,2]. Currently, UAV-assisted communication is being integrated into fifth-generation (5G) networks and is expected to play an even more prominent role in future sixth-generation (6G) networks by supporting ultra-reliable, low-latency, and high-capacity communication. With ongoing advancements in autonomous control, energy efficiency, and spectrum sharing, UAVs are poised to play a critical role in the Internet of Things (IoT), intelligent transportation, smart cities, and emergency response systems, making them a key enabler of next-generation wireless communication [3–5].

Besides UAV communication, full-duplex (FD) transmission is a promising wireless technology that enables a device to transmit and receive signals simultaneously on the same frequency band, effectively doubling spectral efficiency [6,7]. This capability is particularly valuable in IoT and future 6G networks, where massive device connectivity, low latency, and high data rates are essential. FD technology can improve real-time communication among IoT devices and enhance network performance in dense deployments. However, a major challenge in practical FD systems

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is residual self-interference (RSI), which occurs when the transmitted signal leaks into the receiver chain despite self-interference cancellation (SIC) techniques [8,9]. This interference can significantly degrade system performance, especially in scenarios with high transmit power or imperfect hardware [9,10]. Therefore, addressing RSI through advanced signal processing, improved hardware design, and intelligent power control is critical to unlocking the full potential of FD communication in future wireless networks [11].

In the literature, FD transmission is often integrated into UAV systems to leverage the combined advantages of both technologies [12–21]. In particular, outage probability, energy efficiency, and throughput are commonly used performance metrics to analyze the behavior of FD-UAV systems [14–16,18,20,21]. Studies have shown that with effective SIC techniques, FD-UAV systems can outperform conventional half-duplex (HD) UAV systems [15,20,21]. However, when the RSI is sufficiently strong, the performance of FD transmission can fall below that of HD systems [15]. In such cases, the outage probability tends to saturate, and the throughput reaches a ceiling due to the impact of RSI [15,18,21]. To address this issue, employing an optimal transmit power strategy at the UAV can significantly mitigate the effects of RSI, thereby enhancing system performance [13,14,16,19,20]. Moreover, optimizing power allocation not only improves communication quality but also reduces the UAV's power consumption. As a result, the UAV can maintain operation for a longer duration, making it more effective in supporting wireless communication tasks.

Although numerous studies have examined FD-UAV systems, most of them have assumed ideal hardware. In practice, however, residual transceiver hardware impairments (RTHI) are inevitable due to imperfections in analog components such as power amplifiers, mixers, filters, and oscillators. These impairments, including nonlinear distortion, phase noise, in-phase/quadrature imbalance, and amplifier nonlinearities, cannot be completely eliminated, even with advanced calibration and compensation techniques. Recently, some studies have begun to consider the effects of RTHI on the outage probability and throughput performance of FD-UAV systems [15,22,22,23]. These works have shown that neglecting RTHI can lead to inaccurate conclusions about system performance. This is particularly critical in FD-UAV systems, where the simultaneous transmission and reception processes amplify hardware-induced distortions, especially under conditions of high mobility and limited payload capacity [15,22]. As a result, ignoring RTHI can lead to overly optimistic evaluations of system reliability, spectral efficiency, and error performance, potentially resulting in suboptimal system designs.

Therefore, incorporating RTHI into the analytical modeling of FD-UAV systems is essential for developing realistic and robust communication solutions. Moreover, existing studies have not derived important performance metrics such as symbol error probability and achievable rate. Additionally, many of these works employed normalized system configurations, thereby overlooking the influence of practical system parameters. In other words, the effects of several critical parameters in FD-UAV systems remain insufficiently explored.

Motivated by the above issues, in this paper, we present a more accurate and practical analytical framework that captures both the advantages of FD-UAV communication and the adverse effects of RTHI. We analyze the impact of RTHI and RSI on key performance metrics, including symbol error probability (SEP) and achievable rate (AR), aiming to reveal system limitations and support the design of more robust FD-UAV system for future wireless networks. Unlike previous studies that often assume ideal hardware, our model explicitly accounts for both RTHI and RSI, thereby reflecting the real-world signal degradation and interference. In addition to proposing a practical system model, we incorporate realistic parameter values to better capture the actual performance behavior of FD-UAV system in the presence of RTHI and RSI. Notably, the FD-UAV system with RTHI and RSI can be applied in various emerging wireless communication scenarios where high data rates, low latency, and flexible deployment are essential. For instance, in 6G-enabled IoT networks, FD-UAVs can serve as aerial relays or base stations to support massive device connectivity in remote or disaster-stricken areas. Despite the challenges posed by RTHI and RSI, FD-UAVs can significantly enhance spectral efficiency by enabling simultaneous transmission and reception. These systems are also valuable in applications such as real-time surveillance, emergency response, and smart agriculture, where maintaining continuous and reliable communication links is critical. Incorporating RTHI and RSI into the system design allows for more realistic performance evaluation and facilitates the development of robust communication strategies for practical deployment.

It is noteworthy that although [15] also investigates a FD-UAV communication system with RTHI, the scope, methodology, and contributions of our work are distinct. First, the system topology differs: [15] studies a two-way FD system in which both terminals and the UAV relay operate in FD mode, whereas our work considers a one-way link in which the source and destination operate in HD mode and only the UAV relay operates in FD mode. This distinction changes the interference model, the analysis structure, and the resulting performance behavior. Second, the performance metrics are different. While [15] focuses on outage probability (OP) and throughput under RTHI and RSI, we analyze SEP and AR under both separate and joint effects of RTHI and RSI. The SEP directly quantifies symbol-level detection errors, which are critical for modulation and coding design, and the AR provides a tighter bound on practical throughput under impairments, enabling more accurate system evaluation than OP and average throughput alone. Third, [15] does not include an explicit optimization of relay transmit power, whereas we derive a closed-form expression for the optimal FD-UAV transmit power P_V^{Opt} that minimizes SEP or maximizes AR. This yields actionable design guidance for energy-efficient and reliable FD-UAV operation. Finally, our work offers practical insights beyond those in [15], including guidelines for modulation order, carrier frequency, bandwidth, and UAV altitude, and demonstrates scenarios in which FD operation with optimized P_V^{Opt} outperforms HD operation even under severe RTHI and RSI. In summary, our study complements rather than duplicates [15] by introducing an error-rate-oriented analysis, an optimal relay power control solution, and practical design rules for FD-UAV relaying under realistic impairments.

The main contributions of this paper are summarized as follows:

- We develop a practical FD-UAV communication model that explicitly incorporates RTHI and RSI, capturing realistic signal degradation effects often neglected in prior work. The model uses realistic parameter values to more accurately represent the performance of FD-UAV systems under these impairments.
- We derive new analytical expressions for SEP and AR, accounting for both the separate and joint effects of RTHI and RSI. An optimal transmit power control scheme at the UAV is proposed to minimize SEP and maximize AR, mitigating the negative impact of these impairments. The theoretical findings are supported by extensive simulations, confirming the model's accuracy and the importance of considering impairments in system design.
- We analyze system performance across various impairment levels, offering insights into the limitations and robustness of FD-UAV communications. Numerical results indicate that when one impairment is mild, the other has a more pronounced effect, while severe impairments lead to saturation in SEP and AR. The proposed power control effectively reduces SEP, improves AR, and avoids performance saturation at high transmit power. Furthermore, optimal power levels are considerably lower than typical UAV transmit powers, reducing energy consumption. The study also investigates how modulation order, bandwidth, UAV altitude, and operating frequency influence performance, culminating in practical design guidelines for FD-UAV systems under RTHI and RSI conditions.

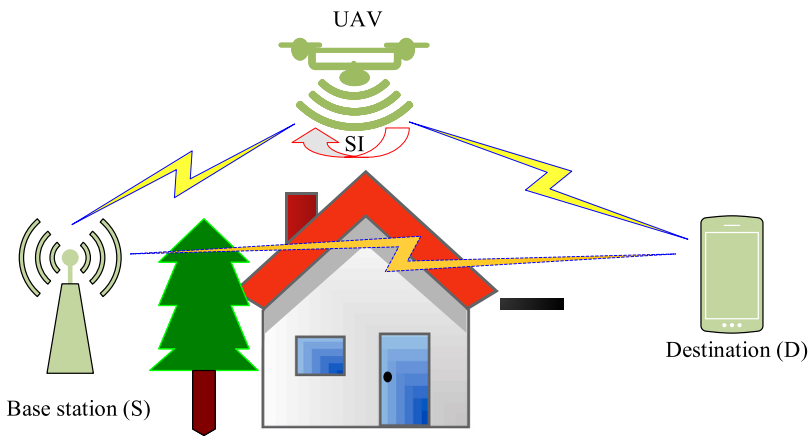

Fig. 1. An illustration of FD-UAV-RTHI systems.

The structure of this paper is as follows. Section 2 introduces the system model and signal description. In Section 3, we conduct a thorough performance analysis centered on the SEP and AR metrics. Section 4 details the proposed optimal transmit power allocation strategy aimed at improving system performance. Numerical results and an in-depth discussion are provided in Section 5. Lastly, Section 6 offers concluding remarks, summarizing the key findings and contributions of this work.

Notation: $\mathbb{E}\{\cdot\}$, $F(\cdot)$, and $f(\cdot)$ denote the expectation operator, cumulative distribution function (CDF), and probability density function (PDF), respectively. $\mathcal{K}_1(\cdot)$ and $\exp(\cdot)$ represent the modified Bessel function of the second kind and the exponential function, respectively. $\Pr\{\cdot\}$ denotes the probability operator, and $\mathcal{CN}(0,\sigma^2)$ represents a circularly symmetric complex Gaussian distribution with zero mean and variance σ^2.

2. System model

Fig. 1 presents the architecture of an FD-UAV system with residual transceiver hardware impairments (UAV-RTHI), which includes two terminal nodes: a base station (S) and a user or destination node (D). Because of the significant distance and possible obstructions between S and D, a UAV is utilized as an aerial full-duplex relay to support the communication link. In this setup, both S and D operate in half-duplex mode, whereas the UAV relays data in full-duplex mode. The base station and destination are each equipped with a single antenna, while the UAV employs two antennas, one for transmitting and the other for receiving. Although it is possible for the UAV to share one antenna for both operations, adopting a dual-antenna configuration substantially enhances the effectiveness of SIC [24,25]. In addition, the UAV uses an amplify-and-forward (AF) scheme.

The signal received at the UAV can be expressed as

$$y_{\mathrm{V}} = h_{\mathrm{SV}}\zeta_{\mathrm{V}}[x_{\mathrm{S}} + n_{\mathrm{S}}^{\mathrm{tx}}] + n_{\mathrm{V}}^{\mathrm{rx}} + \tilde{h}_{\mathrm{VV}}\zeta_{\mathrm{V}}(x_{\mathrm{V}} + n_{\mathrm{V}}^{\mathrm{tx}}) + \tilde{n}_{\mathrm{V}}^{\mathrm{rx}} + z_{\mathrm{V}}, \tag{1}$$

where the term $h_{\mathrm{SV}}\zeta_{\mathrm{V}}[x_{\mathrm{S}} + n_{\mathrm{S}}^{\mathrm{tx}}] + n_{\mathrm{V}}^{\mathrm{rx}}$ denotes the intended signal along with its associated RTHI, $\tilde{h}_{\mathrm{VV}}\zeta_{\mathrm{V}}(x_{\mathrm{V}} + n_{\mathrm{V}}^{\mathrm{tx}}) + \tilde{n}_{\mathrm{V}}^{\mathrm{rx}}$ corresponds to the SI and its respective RTHI, and z_{V} represents the additive white Gaussian noise. Specifically, h_{SV} characterizes the wireless channel from S to the UAV; ζ_{V} models the Doppler-induced channel distortion due to UAV motion and is defined as $\zeta_{\mathrm{V}} = 1/(1+\xi\upsilon)$, where υ is the UAV's velocity and ξ is a constant quantifying the channel sensitivity to speed [26]; x_{S} and x_{V} are the transmitted signals from S and UAV, respectively. The terms $n_{\mathrm{S}}^{\mathrm{tx}}$ and $n_{\mathrm{V}}^{\mathrm{rx}}$ denote the RTHI at the transmitter (S) and receiver (UAV), respectively, over the S–UAV link. The coefficient $\tilde{h}_{\mathrm{VV}}$ signifies the self-interference (SI) path between the transmitting and receiving antennas at UAV. Similarly, $n_{\mathrm{V}}^{\mathrm{tx}}$ and $\tilde{n}_{\mathrm{V}}^{\mathrm{rx}}$ refer to the RTHI at the transmitter and receiver of UAV, associated with the SI channel. Lastly, z_{V} is a complex Gaussian noise term at UAV with variance σ_{V}^2, i.e., $z_{\mathrm{V}} \sim \mathcal{CN}(0,\sigma_{\mathrm{V}}^2)$.

To mitigate the SI component represented by $\tilde{h}_{\mathrm{VV}}\zeta_{\mathrm{V}}(x_{\mathrm{V}}+n_{\mathrm{V}}^{\mathrm{tx}})+\tilde{n}_{\mathrm{V}}^{\mathrm{rx}}$ at the UAV, SIC techniques are employed. These techniques typically combine passive suppression, analog-domain cancellation, and digital signal processing to effectively reduce the power of the SI signal before decoding the desired message [9,15,27]. Despite the application of these cancellation stages, some RSI remains due to hardware imperfections and nonlinearities. This residual component is modeled as a complex Gaussian random variable, denoted as $r_{\mathrm{V}} \sim \mathcal{CN}(0,\tilde{\Omega}_{\mathrm{VV}}\zeta_{\mathrm{V}}^2\Delta_{\mathrm{V}}^2 P_{\mathrm{V}})$, where $\tilde{\Omega}_{\mathrm{VV}}$ is the average SI channel gain, Δ_{V}^2 captures the level of RSI at UAV, and P_{V} is the transmit power of UAV [28,29].

As a result, the overall received signal at the UAV is simplified to

$$y_{\mathrm{V}} = h_{\mathrm{SV}}\zeta_{\mathrm{V}}[x_{\mathrm{S}} + n_{\mathrm{S}}^{\mathrm{tx}}] + n_{\mathrm{V}}^{\mathrm{rx}} + r_{\mathrm{V}} + z_{\mathrm{V}}. \tag{2}$$

This expression highlights the desired signal component from S, along with the RTHI at both the transmitter and receiver, the modeled RSI r_{V}, and the additive Gaussian noise z_{V}. The simplification facilitates further analysis of the system performance under realistic hardware constraints.

It is noteworthy that, based on the definition of the RTHI, the distortion noises introduced at the transmitters are modeled as complex Gaussian random variables [30–33]. Specifically, the distortion at S is given by $n_{\mathrm{S}}^{\mathrm{tx}} \sim \mathcal{CN}(0,(\beta_{\mathrm{S}}^{\mathrm{tx}})^2 P_{\mathrm{S}})$, and the distortion at the UAV is given by $n_{\mathrm{V}}^{\mathrm{tx}} \sim \mathcal{CN}(0,(\beta_{\mathrm{V}}^{\mathrm{tx}})^2 P_{\mathrm{V}})$, where $(\beta_{\mathrm{S}}^{\mathrm{tx}})^2$ and $(\beta_{\mathrm{V}}^{\mathrm{tx}})^2$ represent the RTHI levels at the transmitters of S and UAV, respectively; P_{S} denotes the transmit power of S. Meanwhile, the receiver distortion noises depend not only on the RTHI levels but also on the received signal power, which is influenced by the corresponding channel gains. Accordingly, the distortion at the UAV's receiver due to the S–V link is modeled as $n_{\mathrm{V}}^{\mathrm{rx}} \sim \mathcal{CN}(0,(\beta_{\mathrm{V}}^{\mathrm{rx}})^2 P_{\mathrm{S}}|h_{\mathrm{SV}}|^2\zeta_{\mathrm{V}}^2)$, and the distortion due to the SI link is expressed as $\tilde{n}_{\mathrm{V}}^{\mathrm{rx}} \sim \mathcal{CN}(0,(\beta_{\mathrm{V}}^{\mathrm{rx}})^2 P_{\mathrm{V}}|\tilde{h}_{\mathrm{VV}}|^2\zeta_{\mathrm{V}}^2)$, where $\beta_{\mathrm{V}}^{\mathrm{rx}}$ quantifies the RTHI level at the UAV's receiver [30–32,34,35]. Consequently, we derive the following expected power of the aggregate distortion at the UAV:

$$\mathbb{E}_{n_{\mathrm{S}}^{\mathrm{tx}},n_{\mathrm{V}}^{\mathrm{rx}}}\left\{\left|h_{\mathrm{SV}}\zeta_{\mathrm{V}}n_{\mathrm{S}}^{\mathrm{tx}} + n_{\mathrm{V}}^{\mathrm{rx}}\right|^2\right\} = P_{\mathrm{S}}|h_{\mathrm{SV}}|^2\zeta_{\mathrm{V}}^2\left[(\beta_{\mathrm{S}}^{\mathrm{tx}})^2 + (\beta_{\mathrm{V}}^{\mathrm{rx}})^2\right]. \tag{3}$$

As illustrated in (3), the overall distortion power at the UAV is jointly influenced by multiple parameters: the average transmit power P_S, the instantaneous channel gain $|h_{SV}|^2$ between S and UAV, the Doppler-induced scaling factor ζ_V^2 due to UAV mobility, and the RTHI levels at both the transmitter S and receiver UAV, denoted by $(\beta_S^{tx})^2$ and $(\beta_V^{rx})^2$, respectively. This highlights the critical role of hardware quality and UAV dynamics in determining the effective interference level at the UAV receiver.[1]

Since the UAV employs an AF relaying protocol, its transmitted signal is expressed as

$$x_V = \mathcal{G} y_V, \tag{4}$$

where $\mathcal{G}$ denotes the amplification gain applied by the UAV. This gain is designed to ensure that the UAV transmits at a fixed power level equal to P_V. Mathematically, this constraint is given by

$$\mathbb{E}\left\{|x_V|^2\right\} = \mathcal{G}^2 \mathbb{E}\left\{|y_V|^2\right\} = P_V. \tag{5}$$

Substituting the expression for y_V from (2) and using the statistical properties of the distortion and noise terms, $\mathbb{E}\left\{|y_V|^2\right\}$ is expressed as

$$\begin{aligned}\mathbb{E}\left\{|y_V|^2\right\} &= P_S|h_{SV}|^2\zeta_V^2 + P_S|h_{SV}|^2\zeta_V^2\left[(\beta_S^{tx})^2 + (\beta_V^{rx})^2\right] + \tilde{\Omega}_{VV}\zeta_V^2\Delta_V^2 P_V + \sigma_V^2 \\ &= P_S|h_{SV}|^2\zeta_V^2\left[1 + (\beta_S^{tx})^2 + (\beta_V^{rx})^2\right] + \tilde{\Omega}_{VV}\zeta_V^2\Delta_V^2 P_V + \sigma_V^2.\end{aligned} \tag{6}$$

Thus, the gain $\mathcal{G}$ can be explicitly computed as

$$\mathcal{G} = \sqrt{\frac{P_V}{|h_{SV}|^2\zeta_V^2 P_S\left[1 + (\beta_S^{tx})^2 + (\beta_V^{rx})^2\right] + \tilde{\Omega}_{VV}\zeta_V^2\Delta_V^2 P_V + \sigma_V^2}}. \tag{7}$$

This expression reflects the fact that the amplification gain depends not only on the source transmit power and the S–V channel conditions, but also on the RTHI levels at both S and UAV, the RSI power at the UAV, and the background noise level.

Subsequently, the UAV forwards the amplified signal to D. The received signal at D is given by

$$y_D = h_{VD}\zeta_V(x_V + n_V^{tx}) + n_D^{rx} + z_D, \tag{8}$$

where h_{VD} denotes the channel gain from the UAV to D, and ζ_V again accounts for the Doppler effect due to UAV mobility. The term n_V^{tx} represents the distortion noise at the UAV's transmitter, as previously defined. The receiver distortion at D is modeled as $n_D^{rx} \sim \mathcal{CN}(0, (\beta_D^{rx})^2 P_V|h_{VD}|^2\zeta_V^2)$, where $(\beta_D^{rx})^2$ characterizes the level of RTHI at the destination receiver. Finally, $z_D \sim \mathcal{CN}(0, \sigma_D^2)$ denotes the additive white Gaussian noise at D. This received signal formulation incorporates both RTHI and the UAV's mobility effects, which are critical for accurately analyzing the end-to-end system performance.

By substituting (2) into (4) and then applying the result to (8), the received signal at the destination D can be expressed as

$$\begin{aligned}y_D &= h_{VD}\zeta_V(x_V + n_V^{tx}) + n_D^{rx} + z_D = h_{VD}\zeta_V(\mathcal{G}y_V + n_V^{tx}) + n_D^{rx} + z_D \\ &= h_{VD}\zeta_V\mathcal{G}\left[h_{SV}\zeta_V(x_S + n_S^{tx}) + n_V^{rx} + r_V + z_V\right] + h_{VD}\zeta_V n_V^{tx} + n_D^{rx} + z_D \\ &= h_{VD}\zeta_V\mathcal{G}h_{SV}\zeta_V x_S + h_{VD}\zeta_V\mathcal{G}\left(h_{SV}\zeta_V n_S^{tx} + n_V^{rx}\right) + h_{VD}\zeta_V\mathcal{G}(r_V + z_V) \\ &\quad + h_{VD}\zeta_V n_V^{tx} + n_D^{rx} + z_D.\end{aligned} \tag{9}$$

It is obvious from (9) that the final received signal at D consists of multiple components: the desired signal $\left(h_{VD}\zeta_V\mathcal{G}h_{SV}\zeta_V x_S\right)$, distortions induced by impairments at S and UAV $\left(h_{VD}\zeta_V\mathcal{G}(h_{SV}\zeta_V n_S^{tx} + n_V^{rx})\right)$, interference and noise at UAV $\left(h_{VD}\zeta_V\mathcal{G}(r_V + z_V)\right)$, impairments caused by UAV and D $\left(h_{VD}\zeta_V n_V^{tx} + n_D^{rx}\right)$, and the additive noise at D (z_D).

Consequently, the final signal-to-interference-plus-distortion-and-noise ratio (SIDNR) at D is expressed as

$$\begin{aligned}\gamma_D &= \frac{P_S|h_{SV}|^2|h_{VD}|^2\zeta_V^4\mathcal{G}^2}{|h_{VD}|^2\zeta_V^2\mathcal{G}^2\left[(\beta_S^{tx})^2 P_S|h_{SV}|^2\zeta_V^2 + (\beta_V^{rx})^2 P_S|h_{SV}|^2\zeta_V^2 + \tilde{\Omega}_{VV}\zeta_V^2\Delta_V^2 P_V + \sigma_V^2\right] + \Psi} \\ &= \frac{|h_{SV}|^2|h_{VD}|^2\zeta_V^4\mathcal{G}^2 P_S}{|h_{SV}|^2|h_{VD}|^2\zeta_V^4\mathcal{G}^2 P_S[(\beta_S^{tx})^2 + (\beta_V^{rx})^2] + |h_{VD}|^2\zeta_V^2\mathcal{G}^2(\tilde{\Omega}_{VV}\zeta_V^2\Delta_V^2 P_V + \sigma_V^2) + \Theta},\end{aligned} \tag{10}$$

where $\Psi = (\beta_V^{tx})^2 P_V|h_{VD}|^2\zeta_V^2 + (\beta_D^{rx})^2 P_V|h_{VD}|^2\zeta_V^2 + \sigma_D^2$ and $\Theta = |h_{VD}|^2\zeta_V^2 P_V[(\beta_V^{tx})^2 + (\beta_D^{rx})^2] + \sigma_D^2$.

Then, by using the detailed expression of $\mathcal{G}$ provided in (7), the SIDNR expression at D is formulated as

$$\gamma_D = \frac{|h_{SV}|^2|h_{VD}|^2 P_S P_V}{|h_{SV}|^2|h_{VD}|^2 P_S P_V \beta_{st}^2 + \left[|h_{VD}|^2 P_V(1 + \beta_{VD}^2) + \sigma_D^2/\zeta_V^2\right]\left(\tilde{\Omega}_{VV}\Delta_V^2 P_V + \sigma_V^2/\zeta_V^2\right) + \Xi}, \tag{11}$$

where $\Xi = |h_{SV}|^2 P_S(1 + \beta_{SV}^2)\sigma_D^2/\zeta_V^2$, $\beta_{SV}^2 = (\beta_S^{tx})^2 + (\beta_V^{rx})^2$, $\beta_{VD}^2 = (\beta_V^{tx})^2 + (\beta_D^{rx})^2$, and $\beta_{st}^2 = (\beta_S^{tx})^2 + (\beta_V^{rx})^2 + (\beta_V^{tx})^2 + (\beta_D^{rx})^2 + [(\beta_S^{tx})^2 + (\beta_V^{rx})^2][(\beta_V^{tx})^2 + (\beta_D^{rx})^2]$ represents the combined distortion level due to the RTHI at both the S–UAV and UAV–D links.

The expression in (11) offers important insights into the performance degradation of the system as the levels of RTHI and RSI increase, particularly in UAV communication scenarios.

[1] It should be emphasized that (3) does not represent the received signal itself, which is already provided in (1) and (2) Instead, (3) characterizes the expected power of the aggregate distortion observed at the UAV, originating from both the transmitter-side and receiver-side RTHIs. Specifically, the first term in (3) corresponds to the distortion introduced by the transmitter S, while the second term accounts for the distortion introduced by the receiver UAV. Hence, (3) quantifies the combined distortion power due to hardware impairments, rather than presenting a new received signal expression.

Table 1
Representative values of (a, b) for various modulation schemes.

Modulation schemes	SEP	a	b
BPSK	$Q(\sqrt{2\gamma_D})$	1	2
BFSK	$Q(\sqrt{\gamma_D})$	1	1
QPSK,4-QAM	$\approx 2Q(\sqrt{\gamma_D})$	2	1
N-PSK	$\approx 2Q\left(\sqrt{2\sin^2\left(\frac{\pi}{N}\right)\gamma_D}\right)$	2	$2\sin^2\left(\frac{\pi}{N}\right)$
N-PAM	$\approx \frac{2(N-1)}{N}Q\left(\sqrt{\frac{6}{N^2-1}\gamma_D}\right)$	$\frac{2(N-1)}{N}$	$\frac{6}{N^2-1}$
Non-Rectangular N-QAM	$\approx 4Q\left(\sqrt{\frac{3}{N-1}\gamma_D}\right)$	4	$\frac{3}{N-1}$
Rectangular N-QAM	$\approx \frac{4(\sqrt{N}-1)}{\sqrt{N}}Q\left(\sqrt{\frac{3}{N-1}\gamma_D}\right)$	$\frac{4(\sqrt{N}-1)}{\sqrt{N}}$	$\frac{3}{N-1}$

3. Performance analysis

3.1. Symbol error probability

In general, the SEP can be expressed as [36]

$$\mathrm{SEP} = a\,\mathbb{E}\left\{Q\left(\sqrt{b\gamma_D}\right)\right\} = \frac{a}{\sqrt{2\pi}}\int_0^\infty F_{\gamma_D}\left(\frac{x^2}{b}\right)\exp\left(-\frac{x^2}{2}\right)dx, \tag{12}$$

where $Q(\cdot)$ denotes the Gaussian Q-function. The pair (a, b) corresponds to the modulation scheme, with values specific to different types of modulation (see Table 1). For reference, a complete list of (a, b) values can be found in Tab. 6.1 of [36].

Letting $y = \frac{x^2}{b}$, the SEP expression in (12) can be reformulated as

$$\mathrm{SEP} = \frac{a\sqrt{b}}{2\sqrt{2\pi}}\int_0^\infty \frac{\exp\left(-\frac{by}{2}\right)}{\sqrt{y}}F_{\gamma_D}(y)dy. \tag{13}$$

In the meanwhile, $F_{\gamma_D}(y)$ is calculated as

$$F_{\gamma_D}(y) = \Pr\{\gamma_D < y\}. \tag{14}$$

By employing (11), (14) becomes

$$F_{\gamma_D}(y) = \Pr\left\{\frac{|h_{SV}|^2|h_{VD}|^2P_SP_V}{|h_{SV}|^2|h_{VD}|^2P_SP_V\beta_{st}^2 + \left[|h_{VD}|^2P_V(1+\beta_{VD}^2) + \sigma_D^2/\zeta_V^2\right]\left(\tilde{\Omega}_{VV}\Delta_V^2P_V + \sigma_V^2/\zeta_V^2\right) + \Xi} < y\right\}. \tag{15}$$

From (15), it follows that $F_{\gamma_D}(y) = 1$ when $y \geq \frac{1}{\beta_{st}^2}$. Otherwise, for the case $y < \frac{1}{\beta_{st}^2}$, the expression for $F_{\gamma_D}(y)$ is given by

$$\begin{aligned}
F_{\gamma_D}(y) = 1 - \exp\Bigg(&\frac{-y}{1-y\beta_{st}^2}\Bigg[\frac{\left([G_V^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{VV})]\Delta_V^2P_V+\sigma_V^2/\zeta_V^2\right)(1+\beta_{VD}^2)}{[G_S^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{SV})]P_S} \\
&+\frac{(1+\beta_{SV}^2)\sigma_D^2}{[G_V^{tx}+G_D^{rx}-22.7-26\log(f)-36.7\log(d_{VD})]P_V\zeta_V^2}\Bigg]\Bigg) \\
&\times\sqrt{\frac{4\left([G_V^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{VV})]\Delta_V^2P_V+\sigma_V^2/\zeta_V^2\right)(y+y^2)\sigma_D^2}{[G_S^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{SV})][G_V^{tx}+G_D^{rx}-22.7-26\log(f)-36.7\log(d_{VD})](1-y\beta_{st}^2)^2P_SP_V\zeta_V^2}} \\
&\times\mathcal{K}_1\left(\sqrt{\frac{4\left([G_V^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{VV})]\Delta_V^2P_V+\sigma_V^2/\zeta_V^2\right)(y+y^2)\sigma_D^2}{[G_S^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{SV})][G_V^{tx}+G_D^{rx}-22.7-26\log(f)-36.7\log(d_{VD})](1-y\beta_{st}^2)^2P_SP_V\zeta_V^2}}\right),
\end{aligned} \tag{16}$$

where G_S^{tx} and G_V^{tx} denote the antenna gains of transmitters S and the UAV, respectively; G_V^{rx} and G_D^{rx} denote the antenna gains of receivers at the UAV and D, respectively. Additionally, d_{SV} and d_{VD} represent the distances from S to UAV and from UAV to D, respectively; d_{VV} is the separation between the UAV's transmit and receive antennas; f is the operating frequency; and the constant 22.7 characterizes the environmental propagation loss.[2]

To shorten the obtained formulas, we set

$$\mathcal{X} = \frac{[G_V^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{VV})]\Delta_V^2P_V+\sigma_V^2/\zeta_V^2}{[G_S^{tx}+G_V^{rx}-22.7-26\log(f)-36.7\log(d_{SV})]P_S}, \tag{17}$$

$$\mathcal{Y} = \frac{\sigma_D^2}{[G_V^{tx}+G_D^{rx}-22.7-26\log(f)-36.7\log(d_{VD})]P_V\zeta_V^2}. \tag{18}$$

[2] It is noteworthy that, under the non-line-of-sight (NLoS) scenario, the path loss model is given by $\mathcal{PL} = 22.7 + 26\log(f) + 36.7\log(d_{AB})$, where d_{AB} denotes the distance between nodes A and B [37]. Equivalently, (16) incorporates the path loss model proposed in [37].

Therefore, the final CDF of $\gamma_{\rm D}$ can be expressed as

$$F_{\gamma_{\rm D}}(y)=\begin{cases}1-\exp\Big(\frac{-y}{1-y\beta_{\rm st}^2}[\mathcal{X}(1+\beta_{\rm VD}^2)+\mathcal{Y}(1+\beta_{\rm SV}^2)]\Big)\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Big(\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\Big), \text{ when } y<\frac{1}{\beta_{\rm st}^2},\\ 1, \text{ when } y\geq\frac{1}{\beta_{\rm st}^2}.\end{cases} \tag{19}$$

Substituting (19) into (13), we obtain the SEP of the FD-UAV-RTHI system as stated in the following theorem.

Theorem 1. *The SEP of the FD-UAV-RTHI system is formulated as*

$$\begin{aligned}SEP=&\frac{a\sqrt{b}}{2\sqrt{2\pi}}\Bigg[\sqrt{\frac{2\pi}{b}}-\frac{\pi}{2M\beta_{st}^2}\sum_{m=1}^{M}\sqrt{2\beta_{st}^2(1-\vartheta_m)}\\&\times\exp\Big(-\frac{1+\vartheta_m}{\beta_{st}^2}\Big[\frac{b}{4}+\frac{\mathcal{X}(1+\beta_{VD}^2)+\mathcal{Y}(1+\beta_{SV}^2)}{1-\vartheta_m}\Big]\Big)\\&\times\sqrt{\frac{4\mathcal{X}\mathcal{Y}(1+\vartheta_m)(1+\vartheta_m+2\beta_{st}^2)}{\beta_{st}^4(1-\vartheta_m)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}(1+\vartheta_m)(1+\vartheta_m+2\beta_{st}^2)}{\beta_{st}^4(1-\vartheta_m)^2}}\Bigg)\Bigg],\end{aligned} \tag{20}$$

where $\vartheta_m=\cos\big(\frac{(2m-1)\pi}{2M}\big)$, *$M$ denotes the Chebyshev parameter [31].*

Proof. The comprehensive proofs are presented in Appendix A. □

3.2. Achievable rate

The AR of the FD-UAV-RTHI system, computed in bits per channel use (bpcu), can be expressed as

$$\mathcal{R}=\mathbb{E}\left\{\log_2(1+\gamma_{\rm D})\right\}=\int_0^\infty\log_2(1+y)f_{\gamma_{\rm D}}(y)dy. \tag{21}$$

Starting from (21), and after performing some algebraic simplifications, we obtain

$$\mathcal{R}=\frac{1}{\ln 2}\int_0^\infty\frac{1-F_{\gamma_{\rm D}}(y)}{1+y}dy. \tag{22}$$

Building upon (22), we derive the AR of the FD-UAV-RTHI system as presented in Theorem 2.

Theorem 2. *The AR of the FD-UAV-RTHI system is computed as*

$$\begin{aligned}\mathcal{R}=&\frac{\pi}{M\ln 2}\sum_{m=1}^{M}\frac{\sqrt{1-\vartheta_m^2}}{2\beta_{st}^2+1+\vartheta_m}\exp\Big(\frac{-(1+\vartheta_m)[\mathcal{X}(1+\beta_{VD}^2)+\mathcal{Y}(1+\beta_{SV}^2)]}{2\beta_{st}^2(1-\vartheta_m)}\Big)\\&\times\sqrt{\frac{4\mathcal{X}\mathcal{Y}(1+\vartheta_m)(1+\vartheta_m+2\beta_{st}^2)}{\beta_{st}^4(1-\vartheta_m)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}(1+\vartheta_m)(1+\vartheta_m+2\beta_{st}^2)}{\beta_{st}^4(1-\vartheta_m)^2}}\Bigg).\end{aligned} \tag{23}$$

Proof. From (22), the AR is formulated as

$$\begin{aligned}\mathcal{R}=&\frac{1}{\ln 2}\int_0^{\frac{1}{\beta_{\rm st}^2}}\frac{1}{1+y}\exp\Big(\frac{-y}{1-y\beta_{\rm st}^2}[\mathcal{X}(1+\beta_{\rm VD}^2)+\mathcal{Y}(1+\beta_{\rm SV}^2)]\Big)\\&\times\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\Bigg)dy.\end{aligned} \tag{24}$$

According to [31], (24) is solved as

$$\begin{aligned}\mathcal{R}=&\frac{\pi}{2M\beta_{\rm st}^2\ln 2}\sum_{m=1}^{M}\frac{\sqrt{1-\vartheta_m^2}}{1+\frac{1+\vartheta_m}{2\beta_{\rm st}^2}}\exp\Big(\frac{-\frac{1+\vartheta_m}{2\beta_{\rm st}^2}[\mathcal{X}(1+\beta_{\rm VD}^2)+\mathcal{Y}(1+\beta_{\rm SV}^2)]}{1-\frac{1+\vartheta_m}{2\beta_{\rm st}^2}\beta_{\rm st}^2}\Big)\\&\times\sqrt{\frac{4\mathcal{X}\mathcal{Y}\big[\frac{1+\vartheta_m}{2\beta_{\rm st}^2}+(\frac{1+\vartheta_m}{2\beta_{\rm st}^2})^2\big]}{(1-\frac{1+\vartheta_m}{2\beta_{\rm st}^2}\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}\big[\frac{1+\vartheta_m}{2\beta_{\rm st}^2}+(\frac{1+\vartheta_m}{2\beta_{\rm st}^2})^2\big]}{(1-\frac{1+\vartheta_m}{2\beta_{\rm st}^2}\beta_{\rm st}^2)^2}}\Bigg).\end{aligned} \tag{25}$$

As a result of algebraic rearrangement, Eq. (23) is obtained from (25), thus finalizing the proof. □

4. Power allocation optimization for FD-UAV

In the FD-UAV-RTHI system, the performance reaches a saturation point at high transmit power levels due to the effects of RTHI and RSI. To overcome this limitation and improve system efficiency, we propose an optimal power allocation strategy for the UAV that aims to reduce the SEP and enhance the AR. Since both SEP and AR are derived from the CDF, minimizing the CDF directly leads to lower SEP and higher AR. As

indicated in (19), this optimization involves determining the best transmit power level, P_V. Importantly, the proposed approach is designed not only to alleviate the adverse impact of RSI but also to compensate for the performance degradation caused by RTHI.

The corresponding optimization problem can be stated as:

$$\begin{aligned} &\text{minimize } F_{\gamma_D}(y) \\ &\text{subject to: } P_V > 0,\ P_V \le P_{\max}, \end{aligned} \tag{26}$$

where $F_{\gamma_D}(y)$ is the CDF of the received SIDNR at the destination, P_V is the UAV transmit power to be optimized, and $P_{\max}$ denotes the maximum feasible transmit power imposed by hardware and regulatory constraints. The objective function captures the reliability performance of the system, while the constraints ensure physical feasibility of the solution.

Convexity analysis: By examining the first and second derivatives of $F_{\gamma_D}(y)$ with respect to P_V, it can be shown that $F_{\gamma_D}(y)$ is strictly convex in the feasible domain $P_V \in (0, P_{\max}]$. Specifically, the first derivative $\frac{\partial F_{\gamma_D}(y)}{\partial P_V}$ changes sign only once, and the second derivative $\frac{\partial^2 F_{\gamma_D}(y)}{\partial P_V^2}$ remains positive, which guarantees convexity and thus the existence of a unique global minimizer. This property allows us to employ a differentiation and root-finding approach instead of relying on iterative convex solvers.

Accordingly, the optimal transmit power at the UAV, denoted by P_V^{Opt}, is defined as:

$$P_V^{\text{Opt}} = \arg \min_{0<P_V\le P_{\max}} F_{\gamma_D}(y). \tag{27}$$

To compute this optimum, we solve the Karush–Kuhn–Tucker (KKT) conditions associated with (26). Since the problem is convex, the KKT conditions are both necessary and sufficient for optimality, and solving them reduces to finding the stationary point of $F_{\gamma_D}(y)$. The resulting procedure is summarized in Algorithm 1.

Algorithm 1 Algorithm for Optimal Power Computation

Solve $\frac{\partial F_{\gamma_D}(y)}{\partial P_V} = 0$ to obtain candidate solution P_V^*;
if $0 < P_V^* \le P_{\max}$ **and** $\frac{\partial F_{\gamma_D}(y)}{\partial P_V} < 0$ for $P_V < P_V^*$ **and** $\frac{\partial F_{\gamma_D}(y)}{\partial P_V} > 0$ for $P_V > P_V^*$ **then**
  Output optimal power: $P_V^{\text{Opt}} = P_V^*$;
else
  Output: No feasible optimal power found, $P_V^{\text{Opt}} = \emptyset$;
end if

From Algorithm 1, the optimal transmit power P_V^{Opt} that minimizes the SEP and maximizes the AR of the FD-UAV-RTHI system can be uniquely determined.

Theorem 3. *The closed-form expression of the optimal transmit power P_V^{Opt} for the FD-UAV-RTHI system is given by:*

$$P_V^{Opt} = \sqrt{\frac{\left[G_S^{tx} + G_V^{rx} - 22.7 - 26\log(f) - 36.7\log(d_{SV})\right](1+\beta_{SV}^2)P_S\sigma_D^2}{\left[G_V^{tx} + G_D^{rx} - 22.7 - 26\log(f) - 36.7\log(d_{VD})\right]\left[G_V^{tx} + G_V^{rx} - 22.7 - 26\log(f) - 36.7\log(d_{VV})\right](1+\beta_{VD}^2)\Delta_V^2\zeta_V^2}}. \tag{28}$$

Proof. The comprehensive proofs are presented in Appendix B. □

5. Numerical results and discussions

Building upon the theoretical findings discussed in the earlier sections, this section provides a comprehensive evaluation of the SEP and AR characteristics of the FD-UAV-RTHI system. To validate the theoretical expressions, Monte-Carlo simulations are carried out. Specifically, we employ MATLAB with 10^7 independent channel realizations, where the S-to-UAV and UAV-to-D channels are modeled as independent fading channels in accordance with the considered system model. To underscore the advantages of FD communication, the performance of HD systems is also analyzed for comparison purposes. Additionally, to assess the impact of RTHI, scenarios assuming ideal hardware conditions are considered. In the subsequent figures, the notations "Non-Ideal-FD", "Non-Ideal-HD", "Ideal-FD", and "Ideal-HD" represent the FD-UAV-RTHI, HD-UAV-RTHI, FD-UAV with ideal hardware, and HD-UAV with ideal hardware systems, respectively. It is important to highlight that the performance of the ideal hardware configurations can be derived from the RTHI scenarios by setting $(\beta_S^{tx})^2 = (\beta_V^{tx})^2 = (\beta_V^{rx})^2 = (\beta_D^{rx})^2 = 0$. Similarly, the HD system performance can be extracted from that of the FD systems by setting $\Delta_V^2 = 0$.[3]

In all simulation settings, we assume $(\beta_S^{tx})^2 = (\beta_V^{tx})^2 = (\beta_V^{rx})^2 = (\beta_D^{rx})^2 = \beta^2$, and the transmit powers are set to $P_S = P_V = P_{tx}$ in non-optimized scenarios. In contrast, for the optimized configuration, P_V is determined according to (28). Furthermore, to accurately reflect the system's performance under practical operating conditions, we incorporate the noise power as characterized in [38]. The total noise power is expressed as

$$\sigma_V^2 = \sigma_D^2 = \sigma^2 = \mathcal{N}_y + \mathcal{N}_e + 10\log(\mathcal{B}), \tag{29}$$

where $\mathcal{N}_y$ represents the thermal noise power spectral density, $\mathcal{N}_e$ is the noise figure, and $\mathcal{B}$ denotes the system bandwidth. Accordingly, when $P_S = P_V = P_{tx}$, the average signal-to-noise ratio (SNR) is computed as SNR $= \frac{P_{tx}}{\sigma^2}$. It is important to note that variations in $\mathcal{N}_y$, $\mathcal{N}_e$, or $\mathcal{B}$ directly

[3] It is worth noting that our simulations already provide several comparisons beyond the HD case, including FD systems with ideal hardware versus RTHI, HD versus FD, and FD systems with and without power optimization. These results highlight the significant impact of hardware impairments, RSI, and transmit power optimization on system performance. A comparison with other advanced FD optimization techniques (e.g., beamforming or interference cancellation) is left for future work, as the main focus of this paper is on the joint effect of RTHI, RSI, and power optimization in FD-UAV systems.

Table 2
Simulation parameter settings.

Parameter	Value	Parameter	Value
(x_S, y_S, z_S)	$(0, 0, 5)$	(x_V, y_V, z_V)	$(50, 40, 50 \sim 500)$
(x_D, y_D, z_D)	$(100, 30, 5)$	P_S, P_V	$-10 \sim 45$ dBm
β^2	$0.01 \sim 0.2$	P_V^{Opt}	Calculated from (28)
d_{VV}	0.05 m	Δ_V^2	$-200 \sim -80$ dB
(a, b)	$(1, 2), (2, 1), (3, 1/5)$	ξ	0.05
v	10 m/s	f	$1 \sim 6$ GHz
$\mathcal{N}_e$	10 dB	$\mathcal{B}$	$10 \sim 10000$ MHz
G_S^{tx}, G_V^{rx}, G_V^{tx}, G_D^{rx}	5 dB	$\mathcal{N}_y$	-174 dBm/Hz

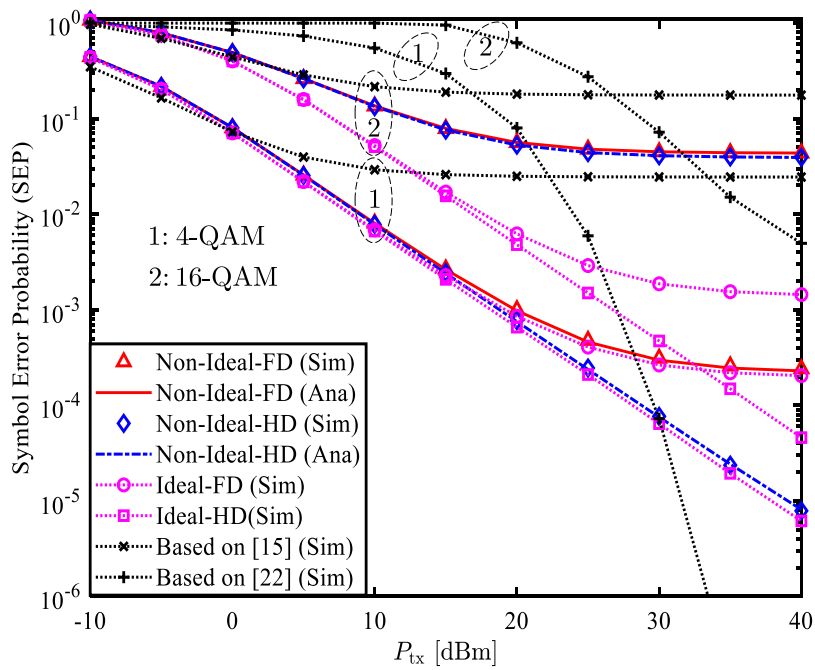

Fig. 2. SEP of the FD-UAV-RTHI system as a function of P_{tx}, with $\beta^2 = 0.01$, $\Delta_V^2 = -130$ dB, $z_V = 80$ m, $f = 5$ GHz, and $\mathcal{B} = 10$ MHz.

affect the effective SNR. For this reason, we present the simulation results in terms of transmit power P_{tx} [dBm] rather than average SNR [dB], as this choice explicitly captures the impact of these noise-related parameters. Nevertheless, the results can equivalently be expressed in terms of SNR without loss of generality.

Additionally, to determine the distances required for calculating the channel gains in (34), (35), and (36), we utilize the three-dimensional spatial coordinates of the nodes. Specifically, let $(x_{N_1}, y_{N_1}, z_{N_1})$ and $(x_{N_2}, y_{N_2}, z_{N_2})$ denote the three-dimensional coordinates of nodes N_1 and N_2, respectively, the Euclidean distance between these two nodes is computed as $d_{N_1N_2} = \sqrt{(x_{N_2} - x_{N_1})^2 + (y_{N_2} - y_{N_1})^2 + (z_{N_2} - z_{N_1})^2}$. The corresponding simulation parameters are summarized in Table 2.

Fig. 2 illustrates the SEP of the FD-UAV-RTHI system as a function of the transmit power P_{tx}, without applying power optimization. This performance is compared to that of the HD-UAV-RTHI system, as well as FD-UAV and HD-UAV systems with ideal hardware. We also compare the SEP of the FD-UAV-RTHI system with those based on [15,22].[4] The simulations are conducted under the following parameter settings: $\beta^2 = 0.01$, $\Delta_V^2 = -130$ dB, $z_V = 80$ m, $f = 5$ GHz, and $\mathcal{B} = 10$ MHz. Two modulation schemes are considered: 4-QAM (with $a = 2$, $b = 1$) and 16-QAM (with $a = 3$, $b = 1/5$). Additional simulation parameters are listed in Table 2. The theoretical SEP for the FD-UAV-RTHI setup is computed using Eq. (20), which is also adapted for the HD-UAV-RTHI scenario by setting $\Delta_V^2 = 0$. The results presented in Fig. 2 confirm the accuracy of the analytical expression proposed in Theorem 1.

As shown in Fig. 2, when 4-QAM is employed, the impact of RTHI on the SEP performance of both FD-UAV-RTHI and HD-UAV-RTHI systems is minimal. This is evident as their SEPs closely match those of systems with ideal hardware. Notably, even in the ideal hardware case, the SEP

[4] To ensure fairness in the comparison, we normalize the system parameters of [15,22] to match those adopted in our work. For example, UAV speed was not considered in [15,22]; however, we include this parameter in their performance metrics for comparison. In addition, we set $M = 16$ reflecting elements in the intelligent reflecting surface (IRS) to obtain the SEP based on [22]. Since FD two-way communication was studied in [15], the SEP reported in [15] is significantly higher than that of the FD-UAV-RTHI system. By contrast, with IRS assistance, the SEP in [22] becomes lower than that of the FD-UAV-RTHI system in the high transmit power regime.

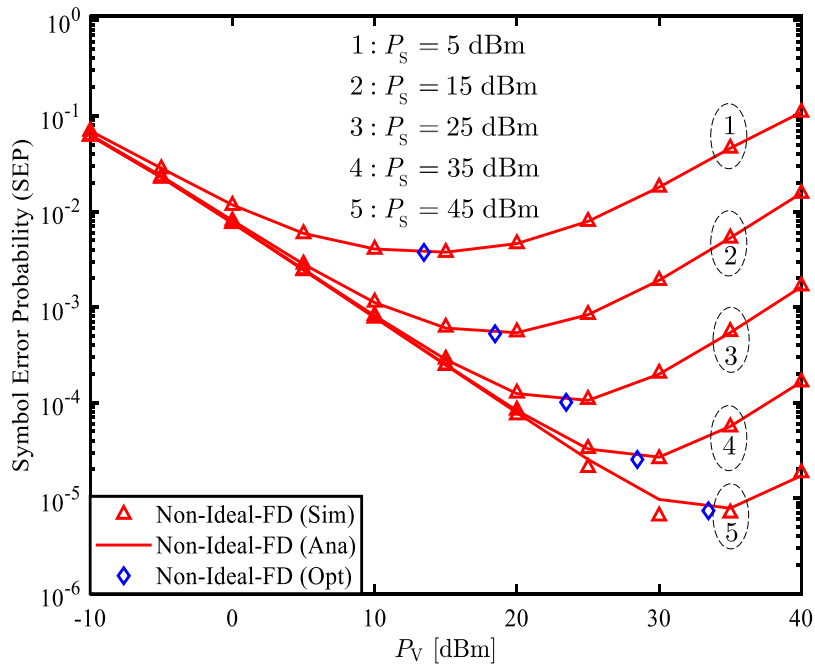

Fig. 3. The SEP of the FD-UAV-RTHI system versus P_V, with and without optimal transmit power, for $\beta^2 = 0.01$, $\Delta_V^2 = -130$ dB, $z_V = 80$ m, $f = 5$ GHz, and $\mathcal{B} = 10$ MHz.

of the FD-UAV system saturates at $P_{tx} = 30$ dBm due to the effect of RSI. Therefore, for low-order modulation such as 4-QAM, the influence of RTHI can be considered negligible. However, for higher-order modulation, such as 16-QAM, the effect of RTHI becomes significant. Specifically, the SEPs of the FD-UAV-RTHI and HD-UAV-RTHI systems are much higher than those of their ideal hardware counterparts. At $P_{tx} = 40$ dBm, the SEPs of the FD-UAV-RTHI and HD-UAV-RTHI systems saturate at approximately 4.1×10^{-2}, whereas the SEPs of the FD-ideal and HD-ideal systems are significantly lower, at 1.5×10^{-3} and 5.3×10^{-5}, respectively. These results highlight the considerable degradation caused by RTHI when higher modulation orders are used. To mitigate this degradation, it is advisable to employ optimal transmit power settings.

Fig. 3 illustrates the SEP of the FD-UAV-RTHI system as a function of P_V for five different values of P_S: 5, 15, 25, 35, and 45 dBm using BPSK (with $a = 1$, $b = 2$). The SEP curve corresponding to the optimal UAV transmit power, derived from (28), is also presented and labeled as "Non-ideal-FD (Opt)" to validate the correctness of the analytical expression in Theorem 3. As shown in Fig. 3, the theoretical optimal transmit power values P_V^{Opt} are 13.5, 18.5, 23.5, 28.5, and 33.5 dBm, corresponding respectively to $P_S = 5, 15, 25, 35$, and 45 dBm. These results demonstrate that when P_S is relatively low (i.e., 5 or 15 dBm), the optimal transmit power at the UAV should be higher than that of the source ($P_V^{Opt} > P_S$) to minimize the SEP. On the other hand, for higher source transmit powers ($P_S = 25, 35$, and 45 dBm), the UAV's optimal transmit power remains lower than the corresponding P_S, while still achieving minimal SEP. Fig. 3 provides important insights into the impact of UAV transmit power optimization on system performance. As observed, when the UAV transmit power is set to the conventional value $P_V = P_S$, the system does not always achieve the lowest SEP. In contrast, optimizing P_V significantly enhances performance. For example, when $P_S = 35$ dBm (Case 4 in Fig. 3), the conventional setting results in $P_V = 35$ dBm, whereas the optimal power is only $P_V^{Opt} = 28.5$ dBm. This clearly demonstrates that the optimal transmit power can be considerably lower than the conventional setting, thereby reducing power consumption without compromising performance. This observation is crucial: it not only ensures SEP minimization but also promotes energy efficiency in UAV operation. In other words, when a low SEP is required, increasing the source's transmit power and adjusting the UAV's transmit power to the optimal value derived from (28) can effectively reduce both the SEP and the UAV's power consumption.

Fig. 4 investigates the effect of RSI (Δ_V^2) on the SEP of the FD-UAV-RTHI system, with and without optimal transmit power, using BPSK modulation. The system parameters are set as $\beta^2 = 0.01$, $z_V = 80$ m, $f = 5$ GHz, and $P_S = 40$ dBm. Various bandwidth values are considered, namely $\mathcal{B} = 10$, 100, 1000, and 10000 MHz. In the case without optimal transmit power, we set $P_V = P_S$. In contrast, for the case with optimal transmit power, P_V^{Opt} is computed using (28). As shown in Fig. 4, when Δ_V^2 is small (e.g., from −200 to −150 dB), its impact on SEP is negligible, as the SEP remains nearly constant in this range. However, as Δ_V^2 increases beyond this range, the SEP increases significantly, even when optimal transmit power is employed. Although using optimal transmit power results in considerably lower SEPs compared to the fixed-power case, the influence of Δ_V^2 becomes prominent as its value increases. For instance, at $\mathcal{B} = 10$ MHz, the SEP rises from 4.9×10^{-6} (with optimal power) and 6.8×10^{-6} (without optimal power) to 4.2×10^{-3} and 4.0×10^{-1}, respectively, as Δ_V^2 increases from −140 to −80 dB. This observation not only highlights the significant impact of Δ_V^2 on system performance but also clearly demonstrates the benefits of employing optimal transmit power, particularly in scenarios where Δ_V^2 is severe. Moreover, increasing the bandwidth $\mathcal{B}$ results in higher noise power (as shown in (29)), which consequently leads to higher SEPs at larger bandwidths.

Fig. 5 evaluates the impact of RTHI, characterized by β^2, on the SEP of the FD-UAV-RTHI system, with and without optimal transmit power. The system parameters are set as $\Delta_V^2 = -110$ dB, $f = 5$ GHz, $\mathcal{B} = 10$ MHz, and $P_S = 40$ dBm. Three UAV altitudes are considered: $z_V = 50$, 150, and 250 m. As shown in Fig. 5, the SEPs with optimal transmit power are significantly lower than those without it, particularly when β^2 is small. Specifically, as β^2 increases from 0.01 to 0.1, the SEPs without optimal power increase from 3.0×10^{-3}, 1.4×10^{-2}, and 3.6×10^{-2} to 2.4×10^{-2}, 4.4×10^{-2}, and 7.4×10^{-2} for $z_V = 50$, 150, and 250 m, respectively. Meanwhile, with optimal transmit power, the SEPs increase from 7.2×10^{-5}, 3.9×10^{-4}, and 1.1×10^{-3} to 1.6×10^{-2}, 1.7×10^{-2}, and 1.9×10^{-2}, respectively. As a result, although optimal transmit power significantly improves

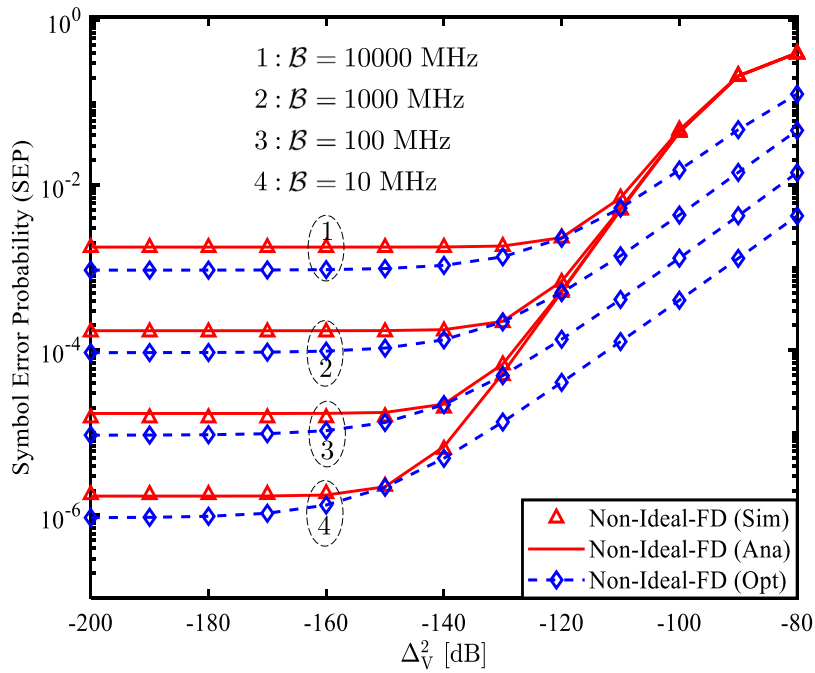

Fig. 4. Effect of Δ_V^2 on the SEP of the FD-UAV-RTHI system, with and without optimal transmit power, for $\beta^2 = 0.01$, $z_V = 80$ m, $f = 5$ GHz, and $P_S = 40$ dBm.

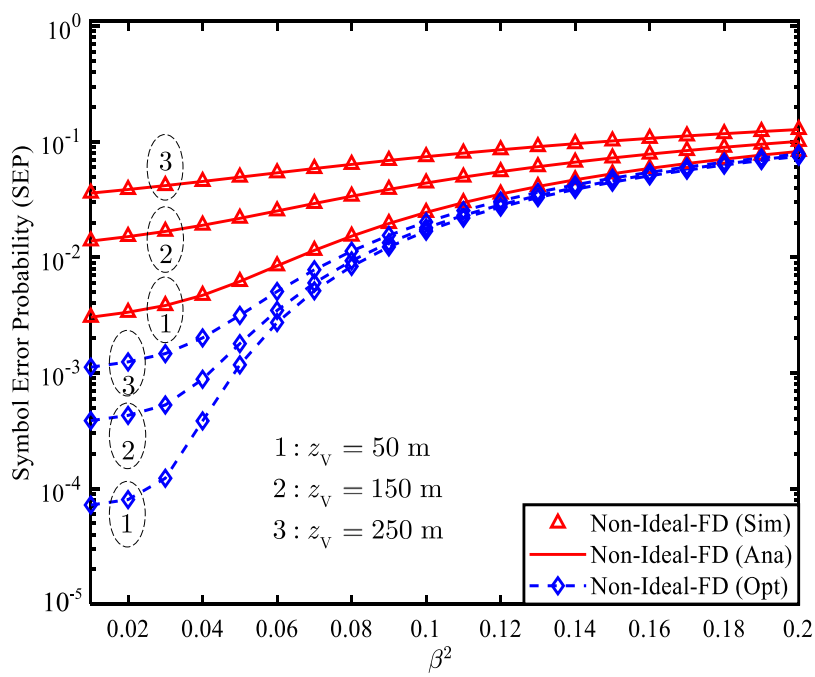

Fig. 5. Impact of RTHI parameter β^2 on the SEP of the FD-UAV-RTHI system, with and without optimal transmit power, for $\Delta_V^2 = -110$ dB, $z_V = 50$, 150, and 250 m, $f = 5$ GHz, $\mathcal{B} = 10$ MHz, and $P_S = 40$ dBm.

performance at low β^2, the SEP degradation rate is steeper compared to the fixed-power case as β^2 increases. This behavior is reasonable: when β^2 is small (e.g., $\beta^2 = 0.01$), the impact of Δ_V^2 dominates, and optimizing transmit power helps reduce SEP effectively. However, when β^2 becomes sufficiently large (e.g., $\beta^2 = 0.2$), its effect overshadows that of Δ_V^2, and thus, even optimal power allocation cannot prevent high SEP.

Fig. 6 illustrates the joint effect of RTHI (β^2) and RSI (Δ_V^2) on the SEP of the FD-UAV-RTHI system, with and without optimal transmit power. As shown in Fig. 6, when both β^2 and Δ_V^2 increase, the SEPs increase rapidly. Specifically, when $\beta^2 = 0.01$ and $\Delta_V^2 = -200$ dB, the SEPs are 1.5×10^{-5} and 7.5×10^{-6} for the cases without and with optimal transmit power, respectively. In contrast, when $\beta^2 = 0.07$ and $\Delta_V^2 = -140$ dB, the corresponding SEPs increase to 5.2×10^{-3} and 5.0×10^{-3}. These results demonstrate that the joint impact of RTHI and RSI is substantial. Therefore, in addition to applying effective SIC techniques in FD communication, further strategies should be developed and deployed to mitigate RTHI in wireless communication systems.

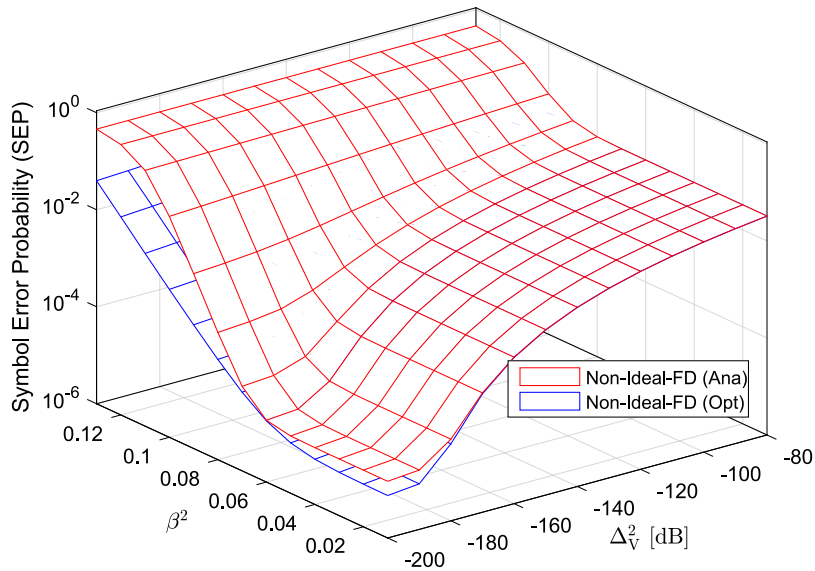

Fig. 6. Joint effect of RTHI and RSI on the SEP of the FD-UAV-RTHI system, with and without optimal transmit power, for $z_V = 250$ m, $f = 5$ GHz, $\mathcal{B} = 10$ MHz, and $P_S = 40$ dBm.

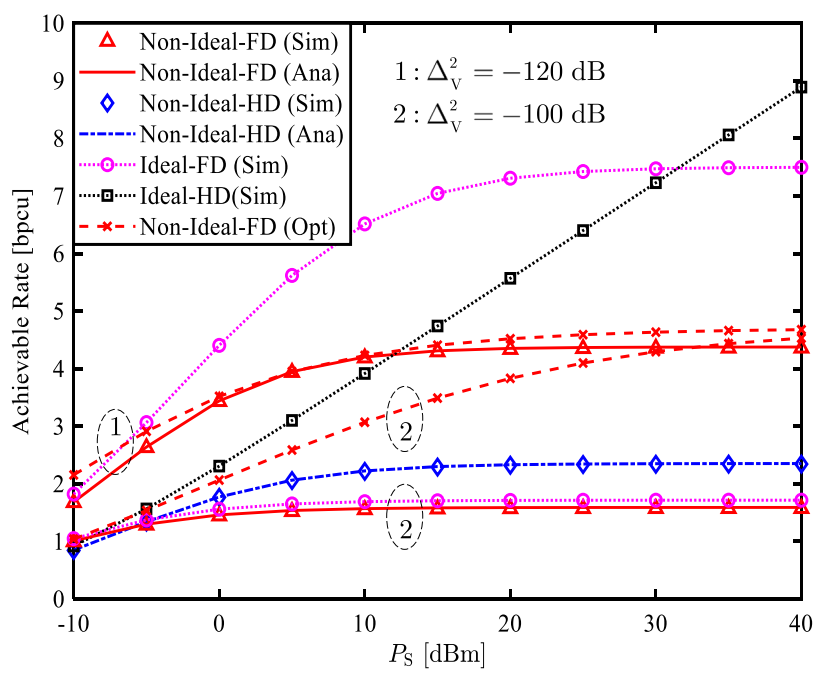

Fig. 7. The AR of the FD-UAV-RTHI system versus P_S, with and without optimal transmit power, for $\beta^2 = 0.01$, $z_V = 90$ m, $f = 2.4$ GHz, and $\mathcal{B} = 10$ MHz.

Fig. 7 illustrates the AR of the FD-UAV-RTHI system as a function of P_S, both with and without optimal transmit power allocation, for $\beta^2 = 0.01$, $z_V = 90$ m, $f = 2.4$ GHz, and $\mathcal{B} = 10$ MHz. Other parameters are identical to those listed in Table 2. Two levels of RSI are considered: $\Delta_V^2 = -120$ dB and $\Delta_V^2 = -100$ dB. The analytical AR is derived from (23). For the HD-UAV-RTHI system, since communication occurs over two time slots, the AR is obtained by multiplying (23) by 1/2. As shown in Fig. 7, when $\Delta_V^2 = -100$ dB (Case 2), the ARs of both the FD-UAV-RTHI and FD-UAV systems with ideal hardware are significantly degraded, lower than the AR of the HD-UAV-RTHI system. This indicates that under high RSI, FD transmission does not provide AR gains over HD transmission. However, even at $\Delta_V^2 = -100$ dB, employing optimal transmit power substantially enhances the AR of the FD-UAV-RTHI system compared to the non-optimized FD-UAV-RTHI system, the FD system with ideal hardware, and the HD-UAV-RTHI system. It is worth noting that under the optimal power allocation, the FD-UAV-RTHI system significantly outperforms the HD benchmark. For instance, as illustrated in Fig. 7, when $\Delta_V^2 = -100$ dB, the FD mode achieves approximately 21% higher average AR compared to the HD mode at $P_S = 10$ dBm, and nearly 100% higher at $P_S = 40$ dBm. These results provide quantitative evidence of the practical advantage of employing FD transmission with optimal power allocation in UAV-assisted networks.

When the RSI is lower, i.e., $\Delta_V^2 = -120$ dB, the FD-UAV-RTHI system achieves nearly double the AR of the HD counterpart. In this case, the ARs with and without optimal transmit power are almost identical, suggesting that transmit power optimization may be unnecessary, thus reducing computational complexity at the UAV. Furthermore, in the ideal hardware scenario, the AR of the FD-UAV system with $\Delta_V^2 = -120$ dB surpasses

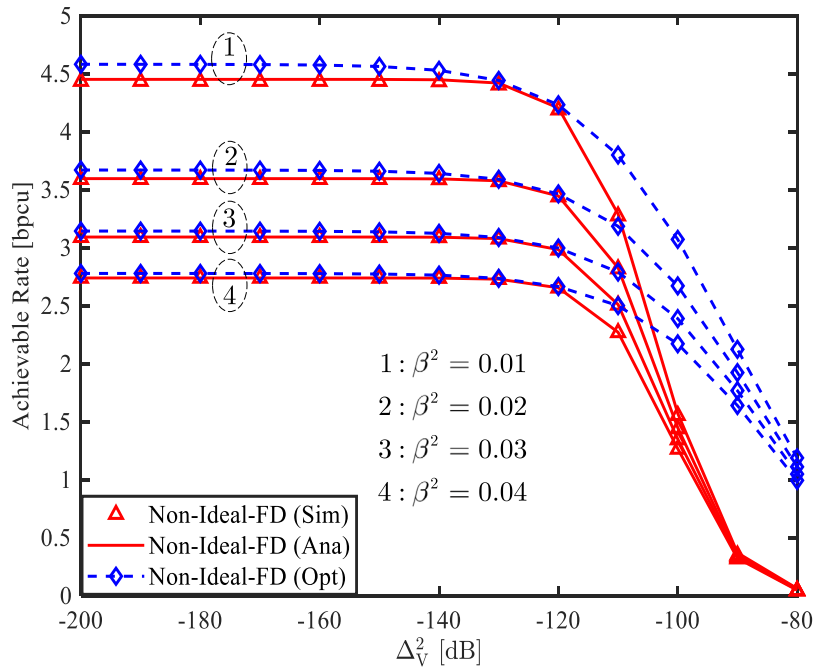

Fig. 8. The effect of RSI level on the AR of the FD-UAV-RTHI system with and without optimal transmit power for $P_S = 10$ dBm, $z_V = 90$ m, $f = 2.4$ GHz, and $\mathcal{B} = 10$ MHz.

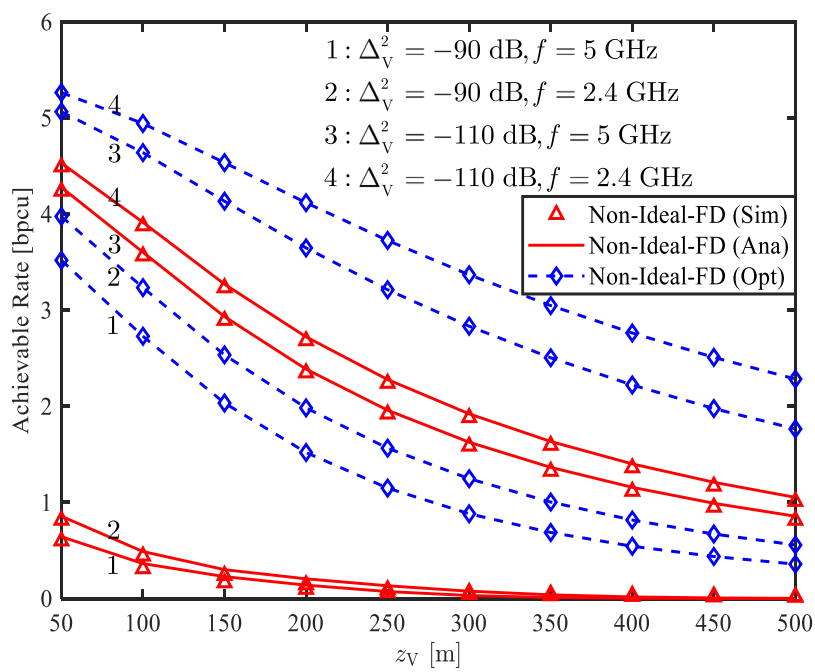

Fig. 9. The AR of the FD-UAV-RTHI system with and without optimal transmit power versus the UAV altitude for $\beta^2 = 0.005$, $P_S = 20$ dBm, and $\mathcal{B} = 10$ MHz.

that of the HD-UAV system when $P_S < 30$ dBm but falls below it when $P_S > 30$ dBm. Therefore, under imperfect SIC, HD transmission may be preferable at high transmit power levels.

Fig. 8 evaluates the effect of the RSI level Δ_V^2 on the AR of the FD-UAV-RTHI system, with and without optimal transmit power, for four levels of RTHI: $\beta^2 = 0.01$, 0.02, 0.03, and 0.04. Similar to the SEP performance, the AR is significantly affected by Δ_V^2 when $\Delta_V^2 > -140$ dB. More specifically, as Δ_V^2 increases from -200 dB to -140 dB, the ARs of the FD-UAV-RTHI system remain nearly unchanged, regardless of whether optimal transmit power is applied. However, for higher RSI values (i.e., Δ_V^2 increasing from -140 dB to -80 dB), the AR rapidly declines, especially in the case without optimal power control. In particular, the AR without optimal transmit power drops to zero at $\Delta_V^2 = -80$ dB, whereas the AR with optimal transmit power remains above 1 bpcu even at this high RSI level. These results further confirm the substantial benefit of employing optimal transmit power in FD-UAV-RTHI systems when the RSI level is sufficiently high. Moreover, consistent with the SEP results, the impact of RSI becomes more pronounced as the RTHI level decreases.

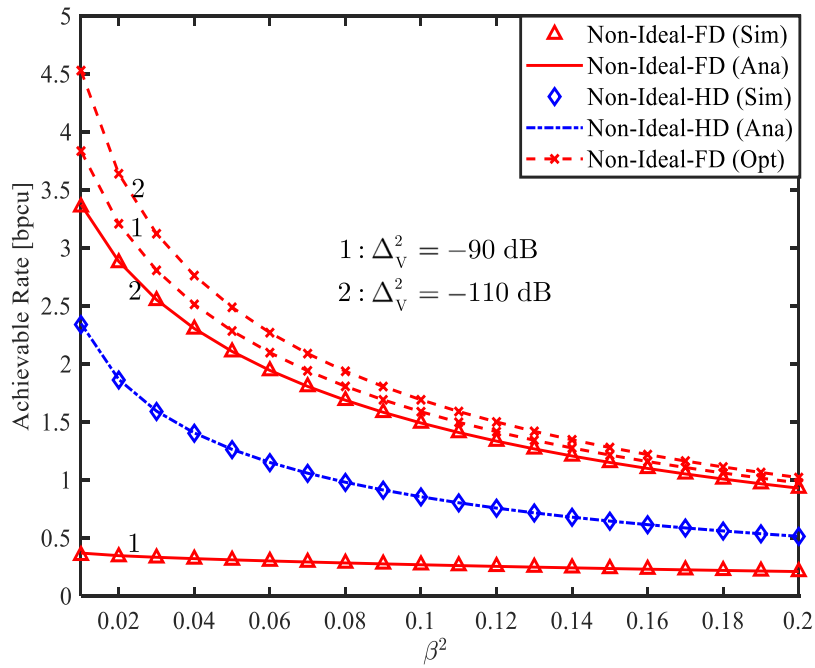

Fig. 10. The impact of RTHI level on the AR of the FD-UAV-RTHI system, with and without optimal transmit power, for $P_S = 20$ dBm, $z_V = 90$ m, $f = 2.4$ GHz, and $\mathcal{B} = 10$ MHz.

Fig. 9 considers the AR of the FD-UAV-RTHI system with and without optimal transmit power as a function of UAV altitude, under varying RSI levels and operation frequencies, i.e., $\Delta_V^2 = -110$ dB, $\Delta_V^2 = -90$ dB, $f = 2.4$ GHz, and $f = 5$ GHz. In other words, Wi-Fi frequency bands are considered in Fig. 9. As observed, when the RSI level is high, i.e., $\Delta_V^2 = -90$ dB (Cases 1 and 2), the ARs of the FD-UAV-RTHI system without optimal transmit power are significantly low, even at a low UAV altitude ($z_V = 50$ m), falling below 1 bpcu. As the altitude increases from 50 m to 500 m, the AR further degrades and eventually drops to zero. In contrast, with optimal transmit power, the system maintains high ARs under the same conditions, reaching approximately 3.5 bpcu and 4 bpcu for $f = 5$ GHz and $f = 2.4$ GHz, respectively, at 50 m altitude. However, these ARs also decline rapidly with increasing altitude, decreasing to about 0.45 bpcu and 0.55 bpcu for $f = 5$ GHz and $f = 2.4$ GHz, respectively, when $z_V = 500$ m. For a lower RSI level, i.e., $\Delta_V^2 = -110$ dB (Cases 3 and 4), the AR trends for both with and without optimal transmit power closely follow the AR curve of the system with $\Delta_V^2 = -90$ dB and optimal power. Notably, even without power optimization, the ARs with $\Delta_V^2 = -110$ dB (for both frequencies) remain higher than those achieved under $\Delta_V^2 = -90$ dB with optimal power control. These results highlight that, in addition to mitigating RTHI and RSI, selecting an appropriate carrier frequency plays a crucial role in preventing performance degradation in FD-UAV-RTHI systems.

Fig. 10 examines the impact of RTHI levels on the AR of the FD-UAV-RTHI system under two different RSI levels, namely $\Delta_V^2 = -90$ dB and $\Delta_V^2 = -110$ dB. As shown in Fig. 10, when $\Delta_V^2 = -90$ dB (Case 1), the AR of the FD-UAV-RTHI system without optimal transmit power is significantly lower than that of the HD-UAV-RTHI system. Therefore, in this case, the FD-UAV-RTHI system must employ the optimal transmit power to outperform its HD counterpart. In contrast, when $\Delta_V^2 = -110$ dB, both the ARs of the FD-UAV-RTHI system with and without optimal transmit power are higher than that of the HD-UAV-RTHI system. In particular, they are nearly two times higher. Moreover, applying the proposed optimal transmit power further enhances the AR of the FD-UAV-RTHI system. These observations suggest that, under the joint effects of RTHI and RSI, the AR of the FD-UAV-RTHI system without power optimization may be either lower or higher than that of the HD system, depending on the interference level. However, when the proposed optimal transmit power is applied, the FD-UAV-RTHI system consistently achieves superior AR performance. This demonstrates that the benefits of full-duplex transmission can be effectively harnessed through power optimization.

6. Conclusion

In this paper, we investigated the SEP and AR performance of the FD-UAV-RTHI system, taking into account the combined effects of self-interference cancellation and hardware imperfections. Specifically, we derived analytical expressions for SEP and AR under both joint and separate impacts of RTHI and RSI. To mitigate the detrimental effects of RTHI and RSI, particularly the RSI introduced by FD transmission, we proposed an optimal transmit power strategy at the UAV to minimize the SEP and maximize the AR. Numerical results validated the significant influence of both RTHI and RSI. Notably, when one impairment was relatively minor, the impact of the other became more pronounced. When both impairments were severe, the SEP and AR quickly reached saturation. In such scenarios, applying the proposed optimal transmit power proved to be an effective solution for reducing SEP, improving AR, and preventing performance saturation in the high transmit power regime. Furthermore, the use of optimal transmit power clearly demonstrated the advantages of full-duplex transmission over half-duplex. Through extensive simulations across diverse scenarios, we thoroughly examined the effects of key system parameters, including modulation order, bandwidth, UAV altitude, and operating frequency, on SEP and AR performance. Based on the observed behaviors, we provided several recommendations for enhancing the performance of FD-UAV-RTHI systems.

CRediT authorship contribution statement

Trong Hai Nguyen: Investigation, Data curation, Conceptualization. **Ba Cao Nguyen:** Software, Project administration, Methodology, Conceptualization. **Tran Manh Hoang:** Visualization, Validation, Methodology, Investigation. **Le The Dung:** Writing – original draft, Supervision, Conceptualization. **Jaydeep Bodwadkar:** Writing – original draft, Methodology, Data curation. **Aditya Kaushik:** Investigation, Conceptualization. **Lam-Thanh Tu:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

Appendix A

This appendix outlines the derivation steps for the SEP expression in Theorem 1, starting with the CDF of $\gamma_{\rm D}$. In particular, from (15), $F_{\gamma_{\rm D}}(y)$ is formulated as

$$\begin{aligned} F_{\gamma_{\rm D}}(y) &= \Pr\{|h_{\rm SV}|^2|h_{\rm VD}|^2 P_{\rm S}P_{\rm V}(1-y\beta_{\rm st}^2) \\ &< \Big[\Big(|h_{\rm VD}|^2 P_{\rm V}(1+\beta_{\rm VD}^2)+\sigma_{\rm D}^2/\zeta_{\rm V}^2\Big)(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V}+\sigma_{\rm V}^2/\zeta_{\rm V}^2)+|h_{\rm SV}|^2 P_{\rm S}(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2\Big]y\}. \end{aligned} \tag{30}$$

As shown in (30), two cases arise: $1-y\beta_{\rm st}^2 \le 0$ and $1-y\beta_{\rm st}^2 > 0$. In the first case, that is, when $1-y\beta_{\rm st}^2 \le 0$ or equivalently $y \ge \frac{1}{\beta_{\rm st}^2}$, Eq. (30) is always satisfied because the term $|h_{\rm SV}|^2|h_{\rm VD}|^2 P_{\rm S}P_{\rm V}(1-y\beta_{\rm st}^2) \le 0$, while the expression $\big[\big(|h_{\rm VD}|^2 P_{\rm V}(1+\beta_{\rm VD}^2)+\sigma_{\rm D}^2/\zeta_{\rm V}^2\big)(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V}+\sigma_{\rm V}^2/\zeta_{\rm V}^2)+|h_{\rm SV}|^2 P_{\rm S}(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2\big]y > 0$. Therefore, the CDF satisfies $F_{\gamma_{\rm D}}(y)=1$ for $y \ge \frac{1}{\beta_{\rm st}^2}$.

In the second case, that is, when $y < \frac{1}{\beta_{\rm st}^2}$, the CDF is calculated as

$$\begin{aligned} F_{\gamma_{\rm D}}(y) = 1 - \int_0^\infty &\Bigg[1 - F_{|h_{\rm SV}|^2}\Bigg(\frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V}+\sigma_{\rm V}^2/\zeta_{\rm V}^2)(1+\beta_{\rm VD}^2)y}{P_{\rm S}(1-y\beta_{\rm st}^2)} \\ &+ \frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V}+\sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{P_{\rm S}P_{\rm V}t(1-y\beta_{\rm st}^2)^2}\Bigg)\Bigg] \\ &\times f_{|h_{\rm VD}|^2}\left(t+\frac{y(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{P_{\rm V}(1-y\beta_{\rm st}^2)}\right)dt, \end{aligned} \tag{31}$$

where the variable is adjusted by defining $|h_{\rm VD}|^2 = t + \frac{y(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{P_{\rm V}(1-y\beta_{\rm st}^2)}$.

It is important to note that the CDF and PDF of the channel gain, specifically $|h_i|^2$ for $i \in \{\rm SV, VD\}$, are provided respectively as detailed in [39], for example,

$$F_{|h_i|^2}(y) = 1 - \exp\Big(-\frac{y}{\Omega_i}\Big), \tag{32}$$

$$f_{|h_i|^2}(y) = \frac{1}{\Omega_i}\exp\Big(-\frac{y}{\Omega_i}\Big), \tag{33}$$

where $\Omega_i = \mathbb{E}\{|h_i|^2\}$.

In this paper, the SEP and AR of the FD-UAV-RTHI system are analyzed under the non-line-of-sight (NLoS) scenario. Therefore, $\Omega_{\rm SV}$ and $\Omega_{\rm VD}$ are respectively expressed as [31,37]

$$\Omega_{\rm SV} = G_{\rm S}^{\rm tx} + G_{\rm V}^{\rm rx} - 22.7 - 26\log(f) - 36.7\log(d_{\rm SV}), \tag{34}$$

$$\Omega_{\rm VD} = G_{\rm V}^{\rm tx} + G_{\rm D}^{\rm rx} - 22.7 - 26\log(f) - 36.7\log(d_{\rm VD}). \tag{35}$$

Notably, since SI can be completely eliminated in ideal line-of-sight (LoS) conditions, the parameter $\tilde{\Omega}_{\rm VV}$ is formulated under the more practical NLoS assumption, which reflects the presence of scattering and multipath effects in the SI channel. Specifically, $\tilde{\Omega}_{\rm VV}$ is also given as [37]

$$\tilde{\Omega}_{\rm VV} = G_{\rm V}^{\rm tx} + G_{\rm V}^{\rm rx} - 22.7 - 26\log(f) - 36.7\log(d_{\rm VV}). \tag{36}$$

Consequently, we obtain

$$F_{|h_{\rm SV}|^2}(y) = 1 - \exp\Big(-\frac{y}{G_{\rm S}^{\rm tx} + G_{\rm V}^{\rm rx} - 22.7 - 26\log(f) - 36.7\log(d_{\rm SV})}\Big), \tag{37}$$

$$\begin{aligned} f_{|h_{\rm VD}|^2}(y) = &\frac{1}{G_{\rm V}^{\rm tx} + G_{\rm D}^{\rm rx} - 22.7 - 26\log(f) - 36.7\log(d_{\rm VD})} \\ &\times \exp\Big(-\frac{y}{G_{\rm V}^{\rm tx} + G_{\rm D}^{\rm rx} - 22.7 - 26\log(f) - 36.7\log(d_{\rm VD})}\Big). \end{aligned} \tag{38}$$

By utilizing (37) and (38), Eq. (31) can be rewritten as

$$F_{\gamma_{\rm D}}(y) = 1 - \int_0^\infty \exp\Big(-\frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V}+\sigma_{\rm V}^2/\zeta_{\rm V}^2)(1+\beta_{\rm VD}^2)y}{\Omega_{\rm SV}P_{\rm S}(1-y\beta_{\rm st}^2)} - \frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V}+\sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{\Omega_{\rm SV}P_{\rm S}P_{\rm V}t(1-y\beta_{\rm st}^2)^2}\Big)$$

$$\times \frac{1}{\Omega_{\rm VD}} \exp\Big(-\frac{t}{\Omega_{\rm VD}} + \frac{y(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{\Omega_{\rm VD} P_{\rm V}(1-y\beta_{\rm st}^2)}\Big)dt$$
$$=1 - \frac{1}{\Omega_{\rm VD}} \exp\Big(-\frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(1+\beta_{\rm VD}^2)y}{\Omega_{\rm SV} P_{\rm S}(1-y\beta_{\rm st}^2)} - \frac{y(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{\Omega_{\rm VD} P_{\rm V}(1-y\beta_{\rm st}^2)}\Big)$$
$$\times \int_0^\infty \exp\Big(-\frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{\Omega_{\rm SV} P_{\rm S} P_{\rm V} t(1-y\beta_{\rm st}^2)^2} - \frac{t}{\Omega_{\rm VD}}\Big)dt. \tag{39}$$

By applying [40, Eq. (3.324.1)], the integral in (39) can be represented as

$$\int_0^\infty \exp\Big(-\frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{\Omega_{\rm SV} P_{\rm S} P_{\rm V} t(1-y\beta_{\rm st}^2)^2} - \frac{t}{\Omega_{\rm VD}}\Big)dt$$
$$=\sqrt{\frac{4\Omega_{\rm VD}(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2}{\Omega_{\rm SV} P_{\rm S} P_{\rm V}\zeta_{\rm V}^2(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2}{\Omega_{\rm SV}\Omega_{\rm VD} P_{\rm S} P_{\rm V}\zeta_{\rm V}^2(1-y\beta_{\rm st}^2)^2}}\Bigg). \tag{40}$$

Substituting (40) into (39), we obtain

$$F_{\gamma_{\rm D}}(y) =1 - \frac{1}{\Omega_{\rm VD}} \exp\Big(-\frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(1+\beta_{\rm VD}^2)y}{\Omega_{\rm SV} P_{\rm S}(1-y\beta_{\rm st}^2)} - \frac{y(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{\Omega_{\rm VD} P_{\rm V}(1-y\beta_{\rm st}^2)}\Big)$$
$$\times \sqrt{\frac{4\Omega_{\rm VD}(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2}{\Omega_{\rm SV} P_{\rm S} P_{\rm V}\zeta_{\rm V}^2(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2}{\Omega_{\rm SV}\Omega_{\rm VD} P_{\rm S} P_{\rm V}\zeta_{\rm V}^2(1-y\beta_{\rm st}^2)^2}}\Bigg)$$
$$=1 - \exp\Big(-\frac{(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(1+\beta_{\rm VD}^2)y}{\Omega_{\rm SV} P_{\rm S}(1-y\beta_{\rm st}^2)} - \frac{y(1+\beta_{\rm SV}^2)\sigma_{\rm D}^2/\zeta_{\rm V}^2}{\Omega_{\rm VD} P_{\rm V}(1-y\beta_{\rm st}^2)}\Big)$$
$$\times \sqrt{\frac{4(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2}{\Omega_{\rm SV}\Omega_{\rm VD} P_{\rm S} P_{\rm V}\zeta_{\rm V}^2(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4(\tilde{\Omega}_{\rm VV}\Delta_{\rm V}^2 P_{\rm V} + \sigma_{\rm V}^2/\zeta_{\rm V}^2)(y+y^2)\sigma_{\rm D}^2}{\Omega_{\rm SV}\Omega_{\rm VD} P_{\rm S} P_{\rm V}\zeta_{\rm V}^2(1-y\beta_{\rm st}^2)^2}}\Bigg). \tag{41}$$

By combining the two cases, i.e., $y < 1/\beta_{\rm st}^2$ and $y \geq 1/\beta_{\rm st}^2$ along with the detailed formulas for $\Omega_{\rm SV}$, $\Omega_{\rm VD}$, and $\tilde{\Omega}_{\rm VV}$ given in (34), (35), and (36), respectively, we obtain (19).

Then, by substituting (19) into (13), the SEP is expressed as

$$\text{SEP} = \frac{a\sqrt{b}}{2\sqrt{2\pi}}\Bigg[\int_0^{\frac{1}{\beta_{\rm st}^2}} \frac{\exp\left(-\frac{by}{2}\right)}{\sqrt{y}}\Bigg[1 - \exp\Big(\frac{-y[\mathcal{X}(1+\beta_{\rm VD}^2) + \mathcal{Y}(1+\beta_{\rm SV}^2)]}{1-y\beta_{\rm st}^2}\Big)$$
$$\times \sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\Bigg)\Bigg]dy + \int_{\frac{1}{\beta_{\rm st}^2}}^\infty \frac{\exp\left(-\frac{by}{2}\right)}{\sqrt{y}}dy\Bigg]. \tag{42}$$

Then, we reorganize (42) as

$$\text{SEP} = \frac{a\sqrt{b}}{2\sqrt{2\pi}}\Bigg[\int_0^\infty \frac{\exp\left(-\frac{by}{2}\right)}{\sqrt{y}}dy - \int_0^{\frac{1}{\beta_{\rm st}^2}} \frac{1}{\sqrt{y}}\exp\Big(-\frac{by}{2} - \frac{y[\mathcal{X}(1+\beta_{\rm VD}^2) + \mathcal{Y}(1+\beta_{\rm SV}^2)]}{1-y\beta_{\rm st}^2}\Big)$$
$$\times \sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\Bigg)dy\Bigg]. \tag{43}$$

By applying [40, Eq. (3.361.2)], the initial integral in (43) is evaluated as

$$\int_0^\infty \frac{\exp\left(-\frac{by}{2}\right)}{\sqrt{y}}dy = \sqrt{\frac{2\pi}{b}}. \tag{44}$$

According to [31], the second integral in (43) is determined as

$$\int_0^{\frac{1}{\beta_{\rm st}^2}} \frac{1}{\sqrt{y}}\exp\Big(-\frac{by}{2} - \frac{y[\mathcal{X}(1+\beta_{\rm VD}^2) + \mathcal{Y}(1+\beta_{\rm SV}^2)]}{1-y\beta_{\rm st}^2}\Big)\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}(y+y^2)}{(1-y\beta_{\rm st}^2)^2}}\Bigg)dy$$
$$= \frac{\pi}{2M\beta_{\rm st}^2}\sum_{m=1}^{M}\frac{\sqrt{1-\vartheta_m^2}}{\sqrt{\frac{1+\vartheta_m}{2\beta_{\rm st}^2}}}\exp\Bigg(-\frac{b(1+\vartheta_m)}{4\beta_{\rm st}^2} - \frac{\frac{[\mathcal{X}(1+\beta_{\rm VD}^2)+\mathcal{Y}(1+\beta_{\rm SV}^2)(1+\vartheta_m)]}{2\beta_{\rm st}^2}}{1-\frac{1+\vartheta_m}{2\beta_{\rm st}^2}\beta_{\rm st}^2}\Bigg)$$
$$\times \sqrt{\frac{4\mathcal{X}\mathcal{Y}\left[\frac{1+\vartheta_m}{2\beta_{\rm st}^2} + (\frac{1+\vartheta_m}{2\beta_{\rm st}^2})^2\right]}{(1-\frac{1+\vartheta_m}{2\beta_{\rm st}^2}\beta_{\rm st}^2)^2}}\mathcal{K}_1\Bigg(\sqrt{\frac{4\mathcal{X}\mathcal{Y}\left[\frac{1+\vartheta_m}{2\beta_{\rm st}^2} + (\frac{1+\vartheta_m}{2\beta_{\rm st}^2})^2\right]}{(1-\frac{1+\vartheta_m}{2\beta_{\rm st}^2}\beta_{\rm st}^2)^2}}\Bigg), \tag{45}$$

where M and ϑ_m are specified in Theorem 1.

Subsequently, by substituting (44) and (45) into (43), the SEP of the FD-UAV-RTHI system is derived as shown in (20). This concludes the proof. □

Appendix B

This appendix outlines the derivation steps for the optimal transmit power expression in Theorem 3. In particular, in high transmit power region, we obtain

$$\frac{4\left([G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VV}})]\Delta_{\mathrm{V}}^{2}P_{\mathrm{V}}+\sigma_{\mathrm{V}}^{2}/\zeta_{\mathrm{V}}^{2}\right)(y+y^{2})\sigma_{\mathrm{D}}^{2}}{[G_{\mathrm{S}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{SV}})][G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{D}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VD}})](1-y\beta_{\mathrm{st}}^{2})^{2}P_{\mathrm{S}}P_{\mathrm{V}}\zeta_{\mathrm{V}}^{2}}\ll 1, \tag{46}$$

$$\left(\frac{-y}{1-y\beta_{\mathrm{st}}^{2}}\left[\frac{\left([G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VV}})]\Delta_{\mathrm{V}}^{2}P_{\mathrm{V}}+\sigma_{\mathrm{V}}^{2}/\zeta_{\mathrm{V}}^{2}\right)(1+\beta_{\mathrm{VD}}^{2})}{[G_{\mathrm{S}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{SV}})]P_{\mathrm{S}}}\right.\right.$$
$$\left.\left.+\frac{(1+\beta_{\mathrm{SV}}^{2})\sigma_{\mathrm{D}}^{2}}{[G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{D}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VD}})]P_{\mathrm{V}}\zeta_{\mathrm{V}}^{2}}\right]\right)\to 0. \tag{47}$$

Next, by applying the approximations $\mathcal{K}_1(A)\approx\frac{1}{A}$ for $A\ll 1$ [41], $F_{\gamma_{\mathrm{D}}}(y)$ given in (16) is approximated as

$$F_{\gamma_{\mathrm{D}}}(y)\approx 1-\exp\left(\frac{-y}{1-y\beta_{\mathrm{st}}^{2}}\left[\frac{\left([G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VV}})]\Delta_{\mathrm{V}}^{2}P_{\mathrm{V}}+\sigma_{\mathrm{V}}^{2}/\zeta_{\mathrm{V}}^{2}\right)(1+\beta_{\mathrm{VD}}^{2})}{[G_{\mathrm{S}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{SV}})]P_{\mathrm{S}}}\right.\right.$$
$$\left.\left.+\frac{(1+\beta_{\mathrm{SV}}^{2})\sigma_{\mathrm{D}}^{2}}{[G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{D}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VD}})]P_{\mathrm{V}}\zeta_{\mathrm{V}}^{2}}\right]\right). \tag{48}$$

Then, applying $\exp(-A)\approx 1-A$ as $A\to 0$ [41], (48) becomes

$$F_{\gamma_{\mathrm{D}}}(y)\approx\frac{y}{1-y\beta_{\mathrm{st}}^{2}}\left[\frac{\left([G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VV}})]\Delta_{\mathrm{V}}^{2}P_{\mathrm{V}}+\sigma_{\mathrm{V}}^{2}/\zeta_{\mathrm{V}}^{2}\right)(1+\beta_{\mathrm{VD}}^{2})}{[G_{\mathrm{S}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{SV}})]P_{\mathrm{S}}}\right.$$
$$\left.+\frac{(1+\beta_{\mathrm{SV}}^{2})\sigma_{\mathrm{D}}^{2}}{[G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{D}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VD}})]P_{\mathrm{V}}\zeta_{\mathrm{V}}^{2}}\right]. \tag{49}$$

At this stage, Algorithm 1 is employed to determine the optimal transmit power at the UAV. Specifically, by differentiating $F_{\gamma_{\mathrm{D}}}(y)$ with respect to P_{V} in (49), we get

$$\begin{aligned}\frac{\partial F_{\gamma_{\mathrm{D}}}(y)}{\partial P_{\mathrm{V}}}=&\frac{\partial}{\partial P_{\mathrm{V}}}\left\{\frac{y}{1-y\beta_{\mathrm{st}}^{2}}\left[\frac{\left([G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VV}})]\Delta_{\mathrm{V}}^{2}P_{\mathrm{V}}+\sigma_{\mathrm{V}}^{2}/\zeta_{\mathrm{V}}^{2}\right)(1+\beta_{\mathrm{VD}}^{2})}{[G_{\mathrm{S}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{SV}})]P_{\mathrm{S}}}\right.\right.\\&\left.\left.+\frac{(1+\beta_{\mathrm{SV}}^{2})\sigma_{\mathrm{D}}^{2}}{[G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{D}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VD}})]P_{\mathrm{V}}\zeta_{\mathrm{V}}^{2}}\right]\right\}\\=&\frac{y}{1-y\beta_{\mathrm{st}}^{2}}\left[\frac{[G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VV}})]\Delta_{\mathrm{V}}^{2}(1+\beta_{\mathrm{VD}}^{2})}{[G_{\mathrm{S}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{SV}})]P_{\mathrm{S}}}\right.\\&\left.-\frac{(1+\beta_{\mathrm{SV}}^{2})\sigma_{\mathrm{D}}^{2}}{[G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{D}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VD}})]P_{\mathrm{V}}^{2}\zeta_{\mathrm{V}}^{2}}\right].\end{aligned} \tag{50}$$

According to (50), there exists a unique point, denoted as P_{V}^{*}, that satisfies the condition $\frac{\partial F_{\gamma_{\mathrm{D}}}(y)}{\partial P_{\mathrm{V}}}=0$, as expressed in the following equation:

$$P_{\mathrm{V}}^{*}=\sqrt{\frac{[G_{\mathrm{S}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{SV}})](1+\beta_{\mathrm{SV}}^{2})P_{\mathrm{S}}\sigma_{\mathrm{D}}^{2}}{[G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{D}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VD}})][G_{\mathrm{V}}^{\mathrm{tx}}+G_{\mathrm{V}}^{\mathrm{rx}}-22.7-26\log(f)-36.7\log(d_{\mathrm{VV}})](1+\beta_{\mathrm{VD}}^{2})\Delta_{\mathrm{V}}^{2}\zeta_{\mathrm{V}}^{2}}}. \tag{51}$$

Given that $\frac{\partial F_{\gamma_{\mathrm{D}}}(y)}{\partial P_{\mathrm{V}}}<0$ for $P_{\mathrm{V}}<P_{\mathrm{V}}^{*}$ and $\frac{\partial F_{\gamma_{\mathrm{D}}}(y)}{\partial P_{\mathrm{V}}}>0$ for $P_{\mathrm{V}}>P_{\mathrm{V}}^{*}$, the second criterion of Algorithm 1 is fulfilled. Therefore, $P_{\mathrm{V}}^{\mathrm{Opt}}=P_{\mathrm{V}}^{*}$ represents the optimal transmit power for the FD-UAV-RTHI system, which completes the proof. □

Data availability

No data was used for the research described in the article.

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